CulinaryState — Group 17

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Draft, 11 November 2026. **AI_DRAFTED / NON_CITABLE / NO_HUMAN_SIGNOFF_D085.** Talk date, duration and author contributions require confirmation. Counts and IDs only; no recipe text. All diagnostic figures retain the labels shown here. This Markdown deck is for author review, not a final export.

What we built

- Recipe-state extraction and deterministic replay.
- A process-effect evidence resource with explicit missingness and provenance.
- Evidence-aware retrieval, answer routing and abstention.
- A narrow ascorbic-acid kinetics pilot, **NOT_VALIDATED / RESEARCH_ONLY**.

The engineering pipeline exists; its construction does not establish the proposal's scientific hypotheses. Source: `FINAL_REPORT_DRAFT_S179_2026-11-11.md` and the cited component reports.

Scope and evaluation

MODEL_ASSISTED_GOLD / NON_CITABLE / NO_HUMAN_SIGNOFF_D085.

D5 frozen benchmark: **74 questions**; S1=12, S2=12, P1/P2/P3/C=10 each, U/X=5 each. The S questions use AI_DRAFTED_GOLD. No independent human gold or inter-rater agreement is claimed. Near-duplicate leakage is not controlled.

Full H2, H4 and later learning lanes are not run before the talk. Step 35 is designed, not run. Sources: D-089, D-090; `registry/h8_scope_cut_register_v1.csv`.

D1: imported evidence and missing conditions

E3_UNVERIFIED / COND_UNAUDITED / BOOTSTRAP_UNVERIFIED / NON_CITABLE.

17,978 ACTIVE rows, **0/40** hand-verified D1 literature rows. All rows are E3. Both temperature and duration are stated on **1,575 rows**; pH is UNKNOWN on all rows. An imported row is not an independently checked primary experiment.

Figure candidate: `reports/s178/evd/fig_evd_coverage_by_family_v2.svg`. Preserve its full-corpus versus sampled denominators in the caption. Source: `sections/S179_D1_evidence_resource_DRAFT.md`.

D3: deterministic engineering, weak gold agreement

The replay corpus has **599,971 non-BBC instruction texts**. **12,046/12,046** selected texts reproduce across process hash seeds. **12,774/599,971** texts fail an invariant. Structural consistency is not recipe-event correctness.

Non-independent gold comparisons use the rule compiler, not a direct Rung A EventGraph scorer. The P1 set has **7 recipes** and HUMAN_ACCEPTED_AI_DRAFTS_NOT_BLIND labels; its handbook requires revision. No independent accuracy claim is made.

Sources: `sections/S179_D3_state_tracking_DRAFT.md`, `reports/s178/corpus/replay_invariants_corpus_v2.json`. Split: near-duplicate leakage not controlled.

H1: rule not met

NON_CITABLE / MODEL_ASSISTED_GOLD / CIRCULAR_SELF_CONSISTENCY_ONLY / NO_HUMAN_SIGNOFF_D085.

Accuracy on **24** S questions: B0 **2/24**, B1=B2=B4 **6/24**. B4 versus each comparator: **4 discordant pairs**, exact two-sided McNemar **p=1.0**. **RULE_NOT_MET**; the small sample does not establish equivalence.

The raw score has H1 NOT_COMPUTED. D-092 uses the separate rung-blind resolution supplement, retained beside it. Sources: `reports/del5/d5_score_v1_D085/d5_h1_after_rung_blind_check_D085.json`; D-092. Split: near-duplicate leakage not controlled.

H2: diagnostic, no direction established

DIAGNOSTIC / BOOTSTRAP_UNVERIFIED / NON_CITABLE / PRIOR_EXPOSURE_REANALYSIS / PROVISIONAL_08A / COND_UNAUDITED / NO_HUMAN_SIGNOFF_D085.

On the provisional Phenol-Explorer study-key split, process-mean loss minus ingredient-mean loss is **0.0992 [−0.0446, 0.2401]**, **362 rows / 20 clusters**. The interval crosses zero. USDA and Bognár each have one lineage group; no study-disjoint estimate exists there. D7 equals D1 on the Phenol-Explorer random predictions.

Full H2 is **DECLARED_NOT_RUN**. The G08-3 lineage audit failed; publication keys are provisional. Source: `sections/S179_H2_and_lineage_statements_DRAFT.md` and 29-diag memos.

D4/H3: a favourable sign test does not pass the rule

DIAGNOSTIC / BOOTSTRAP_UNVERIFIED / NON_CITABLE / NO_HUMAN_SIGNOFF_D085.

B7 is rule-based evidence-sufficiency abstention, **uncalibrated**. The unsupported-claim sign test favours B7 **16–0**, two-sided **p=0.000030517578125**. But on P3, B7 abstains **0/10**, versus B2 **3/10**: **H3 RULE_NOT_MET**.

NEG-0051: gold uses an exact/parent-food boundary; the decision layer admits broad cross-food evidence. The 30-claim audit was NOT_RUN; it could not reverse the failed conjunction. Source: `sections/S179_D4_retrieval_and_abstention_DRAFT.md`; D-092.

Retrieval and routing limits

DIAGNOSTIC / NON_CITABLE / NO_HUMAN_SIGNOFF_D085.

B2 retrieval recall is available on **20 questions**, with plain R@1 **0.2** and R@10 **0.375**. The compatibility-recall values contradict their explanatory note; no compatible-recall claim is made. Source-family stratification is not a train-source-held-out experiment.

Router accuracy is **6/10** with branch gold, **INCOMPLETE_GOLD_BRANCH**; this is not accuracy over all 74 questions. Source: frozen D5 score JSON; D4 section documents the field paths and missing coverage.

AA pilot: conditional diagnostic only

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Frozen Model A: **7 rows, 2 papers, 4 residual degrees of freedom**. Ea **64.40 kJ/mol**. Conditional Student-t 95% interval **[35.57, 93.22] kJ/mol**; the separately reported asymptotic profile interval is narrower and uses a different calibration.

Practical precision: **NOT_ASSESSED**, no locked precision criterion. **NOT_VALIDATED / RESEARCH_ONLY**. Model B sensitivity and separate-context SIM-VAL extraction/precheck are complete. The precheck leaves **0 eligible held-out rows**; no prediction/error metric is emitted. PARAM/domain provenance and production calibration remain incomplete. Sources: D-091; `reports/step10/identifiability_report_v1.md`.

What the results cannot establish

- No independent human-gold H1 result or inter-rater result from AI personas.
- No passed H3 conjunction, calibrated B7 abstention or independent-study H2 generalisation.
- No validated recipe simulator, food-safety advice or whole-system reliability claim.
- No full H4, FlavorTwin, learned Step 35 or later-lane result before the talk.
- Failed 07.A, G08-3 and the handbook revision requirement remain visible.

Original results, deviations and negative findings are retained in the report and release card. Next: resolve the exact learned-model artifact and scientific provenance gaps, then author edits and an actual timed rehearsal. A dated amendment may create a new experiment; it cannot replace the original failed result.

Local completion — 12 November

ENGINEERING_DIAGNOSTIC / NON_CITABLE / NO_HUMAN_SIGNOFF_D085.

- Restored checksum-matched tier-1 data; frozen D5 exposure audit now covers all selected recipes.
- All 79 P/C gold EVD IDs already appear in DEL-2; shared evidence is disclosed.
- Separate strict-v2: 1,000 metadata calls, zero retention figures without exact context.
- 53 physics steps, 141 numeric display fields: zero missing provenance in this sample.
- Public-filtered and local loopback HTTP checks pass. No hosted deployment claimed.
- Exact frozen NER checkpoint still absent; optional v3 checkpoint was not substituted.

Sources: `reports/s179/d5_frozen_split_leakage_v3.md`, `strict_runtime_audit_2026-11-12.json`, `strict_http_smoke_2026-11-12.json`, `optional_model_bundle_check_2026-11-12.json`. Original failed gates and scientific limitations remain unchanged.

Contributions and questions

Author contribution statement pending from Kush and Parv.

References and artifact index: `FINAL_REPORT_DRAFT_S179_2026-11-11.md`, `RELEASE_CARD.md`, `registry/compute_runs_v1.csv`, `registry/negative_results_register_v1.csv`, and round-6 acceptance audit. Before presenting: confirm slot/length, finish report/deck leakage checks, export the approved deck and run it timed.